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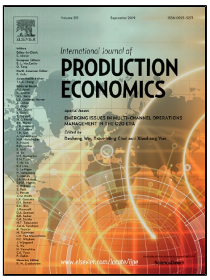

Will Dynamic Pricing Outperform? Theoretical Analysis and Empirical Evidence from O2O On-demand Food Service Market

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Abstract

This study attempts to identify the pricing strategy of three most-commonly used on-demand food service platforms in China based on detailed information of more than 240,000 orders, and further evaluates the impact of pricing strategy on platform performance. We find that some platforms adopt dynamic pricing strategies, while other platforms adopt static pricing strategies. Considering the unique structure of the O2O on-demand food service market in China, we develop a dyadic two-sided market theory and demonstrate that compared with static pricing strategies, dynamic pricing strategies perform better. Consistent with our theory, our empirical analysis also suggests that platforms adopting dynamic pricing strategies have a significantly higher level of demand compared with platforms adopting static strategies. In addition, our results indicate the importance of delivery speed, red envelopes, and ratings in promoting demand.

Keywords: On-demand platforms, Dynamic pricing, Two-sided market, Food service

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1. Introduction

The on-demand economy characterized by online platforms that match consumers with service providers has witnessed explosive growth in recent years (Fowler 2015, Taylor 2018). Different from traditional scheduled service platforms in which customers need to make reservations in advance, on-demand platforms provide service immediately upon receiving the order. Leading examples of such platforms include ride-sharing services (e.g., Uber in the United States, Didi in China, Ola in India), food and/or grocery delivery service (e.g., Doordash, Uber Eats, and Grubhub in the United States, Meituan takeout, Eleme, and Baidu takeout in China). Table 1 lists leading on-demand food service companies around the world in 2018, which shows that most of on-demand food service companies were established after 2000. They have experienced rapid growth and accumulated a large customer base during the past ten years.

The O2O on-demand food service market in China has been booming in recent years. In 2017, China's online food service market was 205.2 billion yuan, and the number of online food ordering customers reached 305 million (Souhu 2019). By June 2017, the top three on-demand food service platforms are Meituan takeout, Eleme, and Baidu takeout, which account for approximately 35.6%, 33.7%, and 16.5% of the total users, respectively (Chyxx 2017). Given the huge market potential of O2O food delivery service in China, it is important to understand the business strategy of on-demand service platforms and how to manage these platforms effectively.

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Table 1. Leading on-demand food service platforms around the world in 2018

Name	Country	Launch time	The number of customers in millions (M)
Meituan takeout	China	2010	More than 300 M
Eleme	China	2009	Approximately 170 M
Koubei	China	2013	Approximately 170 M
Jingdong to Home	China	2015	More than 70 M
Grubhub	U.S.	2004	Approximately 17.2 M
Uber Eats	U.S.	2014	Approximately 8.1 M
DoorDash	U.S.	2013	Approximately 15.5 M
Deliveroo	U.K.	2013	Approximately 6 M
Just Eat	U.K.	2001	Approximately 26.3 M
Deliveryhero	Germany	2011	More than 17 M
Swiggy	India	2014	Approximately 4 M

Note: Baidu takeout in China was not included in the table because it was merged by Eleme in 2017.

In recent years, an emerging literature has started to analyze on-demand markets, and the most commonly examined on-demand markets are two-sided markets, with service providers who possess excess resources (i.e., supply) on one side and customers who need resources (i.e., demand) on the other (Parker and Van Alstyne 2005, Eisenmann et al. 2006). As a result, the central role of the platform is to calibrate supply and demand relative to each other, while guaranteeing relatively high satisfaction to both sides as well as a higher profit for the platform (Riquelme et al. 2015). A typical example of on-demand platform is Uber, which is the world's largest taxi company but owns no vehicles (Goodwin 2015). Uber adjusts the number of drivers through wages and the number of customers through price, to ensure that supply matches demand and customers being picked up and delivered to destinations on time.

Different from the two-sided market structure of on-demand platforms such as

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Uber, the market structure of O2O on-demand food service market in China posits some distinct features with the involvement of merchants, in which customers can order food from many merchants as long as they post their menus on the platform. The platform connects not only merchants and drivers, but also customers and merchants, which we name as a dyadic two-sided market structure. In the first two-sided market, the demand side is the merchants who need delivery services, and the supply side is the drivers who provide such services. In the second two-sided market, the demand side is the customers who need food and the supply side is the merchants who provide them. The platform adjusts the number of consumers, merchants, and drivers through commission fees, with the goals to increase delivery efficiency, expand customer size, attract merchants, and ultimately increase its profits.

A fundamental challenge faced by two-sided markets is the fluctuating supply and demand conditions, which, if not properly balanced, can significantly reduce the effectiveness of service provided and the ability to generate profits (Guda and Subramanian 2017). For instance, a sharp and unexpected increase in demand cannot be fulfilled in time if there are not enough service providers nearby. Compared to regular two-sided markets, a dyadic two-sided market faces more challenges. The platform, which connects customers, merchants, and drivers, not only needs to balance the supply and demand of food between merchants and customers, but also the supply and demand of delivery service between drivers and merchants. A small fluctuation form one side will spillover over to the other two sides, which makes it difficult for the platform to achieve a balance among merchants, customers, and drivers.

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To operate effectively under such market conditions, an increasing number of on-demand platforms adopt a form of dynamic pricing, also known as surging pricing, which indicates that platforms adjust prices in real-time based on algorithms that take into account of factors such as market demand and supply (Riquelme et al. 2015, Zhang et al. 2017). Many factors could impact supply and demand such as extreme weather, promotions, holidays, and weekends, and the essence of dynamic pricing involves changing prices based on these factors. For instance, Didi adopts a surge pricing strategy, which is displayed to customers in the form of a surge multiplier (e.g. 1.5 ×), where the current price is the regular price times the surge multiplier. However, in the O2O on-demand food service market in China, the pricing strategy adopted by each platform is not displayed to the customers. Even though customers can observe fluctuations of delivery fees, it is not clear whether the platform is dynamically adjusting the delivery fee based on real-time supply and demand conditions or simply based on distance or product weight/size.

At present, no clear consensus has been reached in the literature with respect to whether dynamic pricing strategies enhance the performance of on-demand platforms (Guda and Subramanian 2017). Some studies found that surging pricing is helpful when demand for workers exceeds supply, which could attract more workers and meanwhile allocate limited workers to consumers who value them the most (Chen and Sheldon 2015, Hall et al. 2015), while other studies found that surge pricing causes drivers in certain surge price zones to become idle, and it is not able to balance supply and demand (Said 2015, Chen et al 2015). In addition, all the aforementioned studies evaluate the

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effectiveness of dynamic pricing under a two-sided market structure, much less is known regards to the effectiveness of dynamic pricing in a dyadic two-sided market.

To fill the research gap, the objective in this study is to identify the pricing strategy adopted by three major on-demand food service platforms in China (i.e., M, E, and B), and further evaluate its impact on the platform's performance.[1] We first develop a dyadic two-sided market theory and build a theoretical model to demonstrate the advantage of dynamic pricing strategies compared to static pricing strategies. We then obtain a dataset from a restaurant chain that specializes in take-outs, including detailed information of 240,172 orders across all three platforms in a 45-day period from 44 stores. Our empirical results show that the pricing strategy adopted by the platform varies. Some platforms adopt a dynamic pricing strategy (i.e., M and E), whose delivery fees depend not only on distance, but also adjust with factors such as demand, weather, promotions, and holidays, while other platforms (i.e., B) adopt a relatively static pricing strategy, whose delivery fees depend extensively on distance. Consistent with the theoretical model, empirical analysis reveals that dynamic pricing strategies significantly increase the demand.

We contribute to the previous literature in the following aspects. First, we develop a dyadic two-sided market theory based on the unique market structure of the O2O on-demand food service market in China. Based on the theory, we build a theoretical model and demonstrate that dynamic pricing strategies outperform static pricing strategies. In

[1] We use letters to represent the name of the platforms because of the confidentiality of our data.

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addition, we are among the first studies to empirically identify the pricing strategy of major on-demand food service platforms in China. The pricing strategy adopted by each platform is not explicitly shown to the public. By evaluating the sensitivity of delivery fees with respect to factors such as market demand, weather, distance, and holidays, our results shed light on the pricing strategy adopted by each platform, which serves as a prerequisite for the analysis about the effectiveness of pricing strategies. Moreover, we contribute to literature by evaluating the impact of pricing strategy (static vs. dynamic) on the performance of on-demand food service platforms in a dyadic two-sided market structure. Our results have implications about how to effectively improve platform performance by adopting appropriate pricing strategies. Finally, this study complements existing studies that examine complex operations management issues employing a multi-methodological approach (MMA) (Chiu et al. 2019, Chen et al. 2016, Zhao et al. 2016, Choi et al. 2016, Caro and Gallien 2010). By using both analytical modeling and case study research methods, our research results advance knowledge and practice in O2O food service industry.

The rest of this study is organized as follows. Section 2 and 3 present the literature review and the theoretical model. In Section 4, we present data and the associated descriptive statistics. Section 5 identifies the pricing strategy adopted by the three major on-demand food delivery platforms, and evaluates the effects of pricing on platform performance. Conclusions and discussions are presented in Section 6.

2. Literature review

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In this section, we first review studies about dynamic pricing, especially those based on empirical data. We then review literature that evaluates the effectiveness of pricing strategies in the context of on-demand economy, which are classified into two categories, analytical and empirical.

Dynamic pricing has been extensively scrutinized in the revenue management literature under various environments (see Elmaghraby and Keskinocak (2003), Chan et al. (2004) and Talluri and Van Ryzin (2006) for detailed reviews). Some studies examine dynamic pricing in industries with fixed supply and no inventory, such as airlines and hotels (Gallego and van Ryzin 1994, 1997, McGill and van Ryzin 1999). Some work considers the optimal dynamic pricing strategy with inventory decisions (Herbon and Hhmelnitsky 2017, Chew et al. 2014, Gupta et al. 2006). Some studies incorporate strategic consumer responses to pricing (Cachon and Feldman 2015, Papanastasiou and Savva 2016). There are also studies that evaluate dynamic pricing with various demand assumptions (Zhao and Zheng 2000, Zhang et al. 2015). Among studies about pricing strategies, our study is closely related to the literature that compares the performance of dynamic pricing strategies with static pricing strategies using empirical data. For instance, based on a dataset from Airbnb, Li et al. (2016) found evidence that professional service providers significantly outperform non-professionals due to their ability to dynamically change prices. Cheung et al. (2017) developed a pricing strategy based on a dynamic pricing model and then implemented the pricing algorithm at Groupon, a website that sells deals from local merchants. They found the dynamic pricing algorithm significantly improved both daily revenue and

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bookings. Moreover, Zhang et al. (2017) evaluated the causal effects of dynamic pricing through price promotions based on a randomized field experiment with Alibaba Group, and they found that the promotion program doubled the sales of promoted products in the short run.

Platform operations in the context of sharing economy has gained increasing popularity nowadays (Choi et al. 2019, Asian et al. 2019, He et al. 2019). On-demand platforms need to balance supply and demand by adjusting wages paid to service providers and prices charged to consumers. Studies of pricing strategy of on-demand platforms can be classified into two categories. The first category of studies is analytical, which often focus on building pricing models and algorithms (e.g., Taylor 2018, Bai et al. 2018, Riquelme et al. 2015, Cachon et al. 2017, Guda and Subramanian 2017). For instance, both Riquelme et al. (2015) and Cachon et al. (2017) examined whether it would be beneficial for an on-demand service firm to adopt a dynamic pricing strategy, but their results are inconsistent. Riquelme et al. (2015) found that performance under any dynamic pricing strategy cannot exceed that under the optimal static pricing strategy, while Cachon et al. (2017) found that surge pricing achieves nearly optimal profit, and all stakeholders are better off with the application of surge pricing on a platform with self-scheduling capacity.

In addition, both Taylor (2018) and Bai et al. (2018) examined the steady state equilibrium in a queuing model where the supply of independent participating agents depends on wage, and the demand of time-sensitive customer requests depends on price. Taylor (2018) showed that service delays lead to lower prices (to encourage

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participation of customers) and higher wages (to encourage participation of agents) if the platform knows the valuations of consumers and agents; but this may not be the case if the valuations of consumers and opportunity costs of agents are uncertain. Bai et al. (2018) found that it is optimal for the platform to charge a higher price and pay a higher wage when the potential customer demand increases.

Moreover, Guda and Subramanian (2017) analyzed the pricing strategy of a platform facing uncertain demand and supply conditions in adjacent market zones. They showed that when platforms rely on independent workers to serve consumers across different zones, surge pricing is useful in market locations where supply exceeds demand. Gurvich et al. (2019) studied operations in the on-demand economy with self-scheduling service providers. They found that compared to firms that directly dictate when agents work, firms with self-scheduling capacity have lower profits. To reduce compensations paid to agents, on-demand platforms with self-scheduling capacity tend to increase their pool of agents.

The second category of studies focuses on measuring the consequence and effectiveness of dynamic pricing based on empirical data (e.g., Chen and Sheldon 2015, Chen et al. 2015, Hall et al. 2015, Kabra et al. 2016). Some studies find that surging pricing has a positive impact on the performance of on-demand firms. For instance, Hall et al. (2015) analyzed the economics of Uber's surge pricing algorithm. They found evidence that the surge price increases efficiency by increasing the supply of drivers and allocating drivers to consumers who value them the most, so that a ride is almost always five minutes away. Similarly, Chen and Sheldon (2015) examined the

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behavior of Uber drivers based on transactional data associated with 25 million trips, and they showed empirically that dynamic wages significantly increase the supply of rides on the Uber system. Similarly, based on data from a leading ride-hailing marketplace, Kabra et al. (2016) also suggested that passenger incentives and driver incentives are effective in adjusting the number of trips arranged through the market place. In contrast, some studies find that surging pricing negatively affects platform performance. For instance, Chen et al (2015) collected data about pricing, demand, and supply from Uber platform in Manhattan and San Francisco markets over a seven-day period. They found that a surge price has a large and negative impact on demand, which caused drivers in certain surge price zones to become idle and created imbalance between supply and demand.

Overall, even though there are extensive studies about pricing strategies, studies that examine the pricing strategy in the context of O2O on-demand services are still limited and most of them are based on analytical models. Based on first hand data from a leading online restaurant chain in China, we provide first evidence on the efficiency of dynamic pricing in the context of on-demand food service market in China. By employing a multi-methodological approach, we hope that our analysis can serve as a useful guidance for the pricing strategy of on-demand markets.

3. Theoretical framework

The on-demand platform examined by the previous literature is mostly under the framework of a single two-sided market (i.e., Taylor 2018, Bai et al. 2018). For instance,

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as an on-demand driver service platform, Uber matches consumers who need driver service (i.e., demand) and drivers with surplus carrying capacity (i.e., supply). Uber adjusts demand through prices charged to customers, and meanwhile, it adjusts supply by wages paid to drivers. In our case, the market structure becomes complicated with the involvement of merchants. Therefore, our theoretical model is built in a dyadic two-sided market structure that involves on-demand platforms, platform drivers, and merchants. In the theoretical model, we make several important assumptions, which are listed in Table 2.

Table 2. A list of assumptions and reasons in the theoretical model

Assumption1: The platform's revenue consists commissions from merchants, customers, and drivers. **Reasons**: In practice, the platform charges commissions to merchants and customers directly, then give a certain percentage of commissions to drivers as their compensations, which is equivalent to charging a commission to drivers.
Assumption2: The number of merchants, drivers, and customers can be written as increasing functions of their utilities. **Reasons**: This is a common assumption about the number of participants in the two-sided market literature (i.e., Armstrong 2006).
Assumption3: If commissions charged to merchants by the platform is too high, merchants may choose to use other drivers (i.e., non-platform drivers) instead of drivers from the platform (i.e., platform drivers). **Assumption4**: Utility of merchants does not depend on the number of platform drivers. **Reasons**: Merchants are free to choose types of drivers.

Figure 1 presents the cash flow of order handling for an on-demand restaurant service, which can facilitate the understanding of the dyadic two-sided market structure. When a customer places an order from a platform, he pays a total price p to the platform, which includes a commission paid to the platform (i.e., t_c), price paid to the merchant (i.e., p_m) and price for the delivery service (i.e., p_d). Then a proportion of

p_m is deducted by the platform as a commission, denoted by t_m, which is the fee charged by the platform to the merchants for displaying their products on the platform's website and utilizing their delivery personnel, and the rest of p_m, denoted by c_m, is given to the merchants as revenue. Meanwhile, another proportion of p_d is deduced by the platform as a commission, denoted by t_d, which is the fee charged by the platform to drivers, and the rest of p_d is given to the driver as compensation (i.e., c_d). In summary, the platform gets commissions from customers, merchants, and drivers, and its revenue for this order can be expressed as: $\pi = t_m + t_d + t_c$.

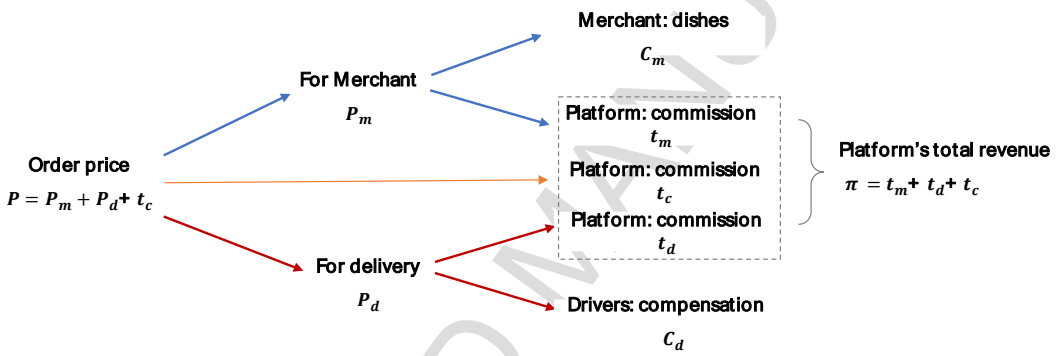

Figure 1. Cashflow of the on-demand food service market

Figure 2 displays the dyadic two-sided market structure in our study. Panel A and panel B represent two-sided market A and B, respectively. In the first two-sided market (i.e., Panel A), the platform connects merchants and drivers. Merchants are the demand side of the delivery service. By adjusting commission t_m, platform adjusts the number of merchants using its delivery service. Specifically, if the commission is set too high, merchants may choose to use non-platform drivers instead of the delivery service offered by the platform (i.e., platform drivers). On the other hand, the number of platform drivers is the supply side of delivery service, which can be adjusted by

commission t_d. A lower commission may encourage the participation of drivers. Overall, platform balances supply and demand of the delivery service in the market by adjusting t_m and t_d.

In the second two-sided market (i.e., Panel B), the platform connects merchants and customers. The supply side of the product/food (i.e., the number of merchants) can be adjusted by commission t_m, while the demand side (i.e., the number of consumers) can be adjusted by commission t_c. By adjusting t_m and t_c, the platform balances the supply and demand of the product/food in the market.

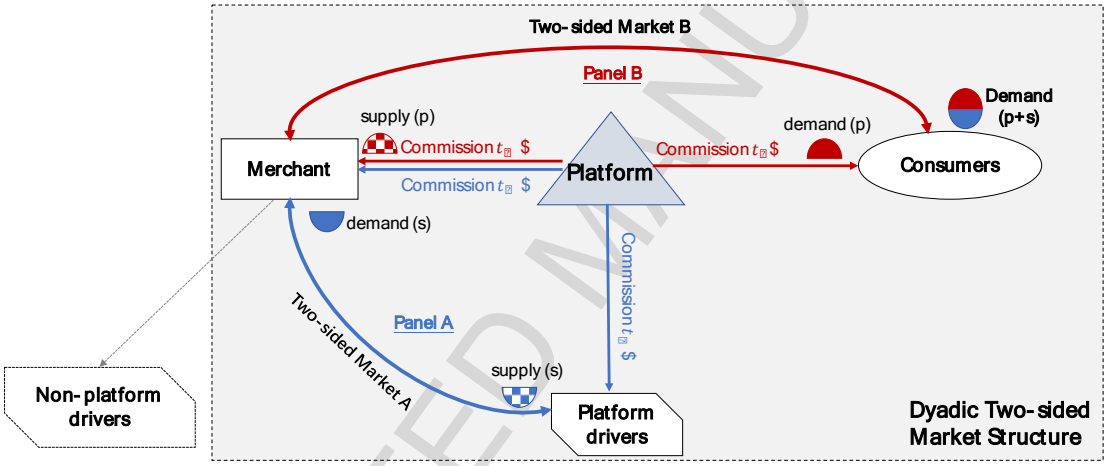

Figure 2. A dyadic two-sided market structure

We assume that the number of consumers, merchants, and drivers are n_c, n_m, and n_d, respectively. The platform can adopt either dynamic pricing strategy or fixed pricing strategy when determining the delivery fee p_d in panel A. Under dynamic pricing strategy, p_d can be written as a function of the number of consumers, while under static pricing strategy, it is a constant. Then, we formulate the utility function for each side as follows.

Consumer's utility function can be defined as: $u_c = \alpha_c n_m - p$, where

$p = c_m + c_d + t_m + t_d + t_c = p_m + p_d + t_c$. Thus, we have

$$t_c = \alpha_c n_m - (c_m + c_d + t_m + t_d) - u_c. \tag{1}$$

Merchant's utility function is $u_m = \alpha_m n_c + (p_m - t_m)$, and then we have

$$t_m = \alpha_m n_c + p_m - u_m. \tag{2}$$

Driver's utility function is $u_d = \alpha_d n_m + (p_d(n_c) - t_d)$, and we can get:

$$t_d = \alpha_d n_m + p_d(n_c) - u_d. \tag{3}$$

Therefore, the platform's profit can be expressed as:

$$\pi = n_m t_m + n_d t_d + n_c t_c. \tag{4}$$

According to Armstrong (2006), we assume the number of each side (n_i) who join the platform follows a form of increasing function, such as $n_i = e^{\lambda_i u_i}$, where $i = c, m, d$. Thus, we have $\frac{dn_i}{du_i} = \lambda_i n_i$.

According to the first-order optimality condition of π with respect to u_m and let $d\pi / du_m = 0$, we have $\lambda_m n_m \left[t_m + \alpha_d (n_d - n_c) + n_c \alpha_c \right] - (n_m - n_c) = 0$. Therefore, we can get the optimal commission charged to merchants as below:

$$t_m^* = \frac{n_m - n_c}{\lambda_m n_m} - \alpha_d (n_d - n_c) + n_c \alpha_c. \tag{5}$$

Similarly, the first-order optimality condition of π with respect to u_d is:

$$\frac{d\pi}{du_d} = \frac{dn_d}{du_d} t_d - (n_d - n_c) = \lambda_d n_d t_d - (n_d - n_c) = 0, \tag{6}$$

and then the optimal commission charged to the drivers t_d^* can be expressed as below:

$$t_d^* = \frac{n_d - n_c}{\lambda_d n_d}. \tag{7}$$

Proposition 1. $t_d^* \geq 0$ for $n_d > n_c$, and $t_d^* < 0$ for $n_d < n_c$.

Proposition 1 indicates that under the circumstance of sufficient delivery capacity, where the number of drivers is larger than the number of consumers, platforms will charge a positive commission to the drivers. In contrast, under the circumstance of insufficient delivery capacity, where the number of drivers is smaller than the number of consumers, platforms pay a positive fee to drivers in addition to the compensation to attract more drivers.

According to Eq. (4), from the first-order derivative condition of π with respect to u_c, we have

$$\frac{d\pi}{du_c} = \lambda_c n_c \left[(n_m - n_c)\alpha_m + (n_d - n_c)\frac{dp_d}{dn_c} - t_c \right] - n_c = 0 . \tag{8}$$

Thus, the optimal commission charged to the consumers t_c^* can be expressed as:

$$t_c^* = \frac{1}{\lambda_c} - (n_m - n_c)\alpha_m - (n_d - n_c)\cdot\frac{dp_d}{dn_c} . \tag{9}$$

Substituting t_m^*, t_d^*, and t_c^* to equation (4), we can get:

$$\pi^* = n_m t_m{}^* + n_d t_d{}^* + n_c t_c{}^* = n_m \left[\frac{n_m - n_c}{\lambda_m n_m} - \alpha_d (n_d - n_c) + n_c \alpha_c \right]$$
$$+ n_d \left(\frac{n_d - n_c}{\lambda_d n_d} \right) + n_c \left[\frac{1}{\lambda_c} - (n_m - n_c)\alpha_m - (n_d - n_c)\cdot\frac{dp_d}{dn_c} \right] . \tag{10}$$

For notation convenience, we define parameter $t_c^0 = \frac{1}{\lambda_c} - (n_m - n_c)\alpha_m$ to denote the fixed price charged to consumers, i.e., $dp_d / dn_c = 0$. On the contrary, we formulate dynamic pricing mechanism as $dp_d / dn_c \neq 0$. Specifically, according to Proposition 1, we have $\frac{dp_d}{dn_c} < 0$ for $n_d > n_c$, and $\frac{dp_d}{dn_c} > 0$ for $n_d < n_c$. When there is sufficient delivery capacity (i.e., $n_d > n_c$), an increasing number of consumers will lower the

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delivery fee because drivers will compete for orders by accepting lower compensations (i.e., $\frac{dp_d}{dn_c} < 0$). In contrast, when there is insufficient delivery capacity (i.e., $n_d < n_c$), an increasing number of consumers will push the platform to offer higher delivery fees to attract more drivers (i.e., $\frac{dp_d}{dn_c} > 0$). Accordingly, $(n_d - n_c) \cdot \frac{dp_d}{dn_c} < 0$ always holds for $n_d > n_c$ and $n_d < n_c$, i.e., $t_c^* > t_c^0$ Thus, we have the following proposition.

Proposition 2. $\pi^*\left(t_m^*, t_d^*, t_c^*\right) > \pi^*\left(t_m^*, t_d^*, t_c^0\right)$.

Proposition 2 implies that the platform should adopt dynamic pricing strategies to achieve higher profits whether he has sufficient delivery capacity or not. In the next section, to confirm the theoretical model, we collect data from a restaurant chain and empirically test if the propositions hold in practice.

4. Data

4.1 Variables

Our data are obtained from a restaurant chain that operates only on take-away orders across all three platforms (i.e., M, E, and B). We collect data of 240,172 orders from 44 stores during the time period between May 1st and June 9th in 2017, each of which contains detailed order information such as delivery fees, order content (i.e., price and promotions), and the time the order was placed and delivered.

Table 3 presents the definitions of the variables in the analysis. Their descriptive statistics and correlation matrix are summarized in Table 4 and 5, respectively. Panel A lists the variables that are used to determine whether the platform employs a dynamic

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pricing strategy, and panel B lists the variables that are used to analyze the impact of dynamic pricing on platform performance (i.e., demand).

The dependent variable in panel A is the delivery fee per order, which could be affected by the following factors. First, distance directly affects delivery fees because longer distance takes more time to deliver. Since distance has the most straightforward impact on delivery fees, delivery fees of platforms with static pricing strategies may depend on the distance to a larger extent compared to platforms with dynamic pricing strategies.

Second, delivery fees could be affected by holidays and weekends on both directions. On one hand, the number of delivery personnel may be reduced due to holidays and weekends, resulting in reduced delivery capacity. To maintain the delivery capacity at a stable level, the platform may raise delivery fees. On the other hand, our data indicate that the number of orders is significantly lower during holidays and weekends. From this perspective, delivery fees may be adjusted downward by the platform because the demand for the delivery service is reduced.

Third, the distribution of the number of orders in one day is a double peak curve, as shown later in Figure 4, and a high concentration of orders may lead to insufficient delivery capacity. To alleviate delivery pressure and reduce the probability of delayed orders, the platform may adjust the delivery fee upward during peak hours to attract more drivers. Therefore, to capture the impacts of peak hours on delivery fees among different platforms, we include the top two peak hours as shows by our data, which are from 11 AM to 12 PM (i.e., Peak1) and 12 PM to 1 PM (i.e., Peak2).

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Fourth, the salary for many delivery personnel is completely based on the number of orders delivered (i.e., no base salary). For these deliverers, if the delivery fee is the same, they will choose the order with smaller volume and weight so as to carry more orders during one trip. Therefore, to ensure the timeliness and service quality of high price orders, which typically have higher weights, the platform may offer higher delivery fees for these orders.

Finally, we consider the impact of rainy weather on delivery fees. Road congestion caused by the rain will reduce the delivery efficiency. In addition, rainy weather causes more consumers ordering online. Thus, the platform may increase the delivery fee to reduce the delivery pressure during rainy days.

Table 4 shows that among 240,172 orders, the average delivery fee is 4.24 yuan and the average distance is 1.56 kilometers. Approximately 29% of the orders are placed during holidays and weekends. Additionally, more than 50% of the orders are placed during peak hours from 11 AM to 1 PM. In specific, 34% of the orders are placed from 11 AM to 12 PM, and 18% are placed from 12 PM to 1 PM. The average order price is 40.58 yuan, and 20% of the orders are placed during rainy weathers.

Table 3 also lists the variables used in panel B. The dependent variable in panel B is the number of orders for each platform for each hour from 8 AM to 9 PM, which could be affected by the following factors. First, delivery speed is an important factor when consumers choose food service platforms. If a consumer experiences delays frequently from one platform, or feels the delivery speed is not satisfactory, it imposes no cost for the consumer to switch to another platform. Therefore, we expect that

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increased delivery speed positively affects demand. In our study, average delivery speed is the ratio of total delivery time to total delivery distance per hour.

Second, demand may be affected by delivery fees charged by the platform, and a high delivery fee may reduce the demand of the delivery service. In our study, average delivery fees for each platform is the ratio of total delivery fees to total delivery distance per hour.

Third, we include red envelopes and platform discounts as explanatory factors affecting demand. Red envelopes and discounts are both promotion strategies adopted by platforms, with the purpose to increase demand. Red envelopes are monetary gifts provided by the platform, and it can be used in any merchants listed in the platform, while discounts are usually targeted for a specific merchant. In the current study, red envelopes and platform discounts are calculated as the average red envelopes and discounts received by each order.

Fourth, we include platform ratings as an additional explanatory variable, expecting that a higher rating will increase platform demand. Finally, holidays, weekends, and rainy weathers are included as dummy variables, which are all expected to affect demand. As described above, holidays and weekends may reduce the demand for online food services, while rainy weather is expected to increase the demand.

Table 4 presents descriptive statistics of panel B. We find that, on average, the hourly demand is 153, the delivery speed is 8.57 minutes, and the delivery fee is 2.78 yuan. Additionally, on average, each order receives a red envelope of 0.63 yuan and a discount of 1.33 yuan. The average rating is 0.43. The correlation matrix of both panel

A and panel B is displayed in Table 5. Panel A shows that distance, Peak1, order price, and rain are positively correlated with delivery fees, while holiday and weekends are negatively correlated with delivery fees. Additionally, from Panel B, we find red envelopes, discounts, and ratings are positively correlated with demand. We do not observe high correlation coefficients (i.e., larger than 0.7), and multicollinearity does not affect the unbiasedness of estimation result, thus multicollinearity should not bias our estimation results.

Table 3. Summary of variables

Variable type	Variable	Variable definition
Panel A		
Dependent variable	Delivery fees	Delivery fees
Independent variables	Distance	Distance between restaurant and customer
	Holiday	Whether it is a holiday, 1 indicates yes
	Weekend	Whether it is a weekend, 1 indicates yes
	Peak1	Whether the order is delivered during peak hour (11 AM to 12 PM), 1 indicates yes
	Peak2	Whether the order is delivered during peak hour (12 PM to 1 PM), 1 indicates yes
	Price	Total price of the order
	Rain	Whether it is a rainy day, 1 indicates yes
Panel B		
Dependent variable	Demand	Number of orders
Independent variables	Speed	Average delivery speed
	Delivery fees	Average delivery fees
	Red envelopes	Average red envelopes
	Discounts	Average discounts
	Ratings	Overall rating of the platform
	Holiday	Whether it is a holiday, 1 indicates yes
	Weekend	Whether it is a weekend, 1 indicates yes
	Rain	Whether it is a rainy day, 1 indicates yes

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Table 4. Descriptive statistics

Variable	Number of Observations	Mean	Std. Dev.	Min	Max
Panel A					
Delivery fees	240172	4.24	1.13	1	29
Distance	240172	1.56	0.81	0	12.43
Holiday	240172	0.06	0.25	0	1
Weekend	240172	0.23	0.42	0	1
Peak1	240172	0.34	0.47	0	1
Peak2	240172	0.18	0.39	0	1
Order price	240172	40.58	26.07	9.5	1521
Rain	240172	0.20	0.4	0	1
Panel B					
Demand	1564	153	261	1	1684
Speed	1564	8.57	8.77	0	86
Delivery fees	1564	2.78	0.98	0	14
Red envelopes	1564	0.63	0.97	0	8
Discounts	1564	1.33	1.75	0	16
Ratings	1564	0.43	0.75	0	5
Holiday	1564	0.09	0.28	0	1
Weekend	1564	0.27	0.44	0	1
Rain	1564	0.12	0.32	0	1

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Table 5. Correlation matrix

Panel A	Delivery fees	Distance	Holiday	Weekend	Peak1	Peak2	Order price	Rain
Delivery fees	1							
Distance	0.242***	1						
Holiday	-0.010***	0.015***	1					
Weekend	-0.016***	0.020***	0.062***	1				
Peak1	0.012***	-0.032***	-0.050***	-0.071***	1			
Peak2	0.000	-0.015***	0.015***	0.034***	-0.337***	1		
Order price	0.164***	0.034***	-0.002	0.005***	0.018***	-0.007***	1	
Rain	0.046***	0.001	0.276***	-0.123***	-0.020***	-0.008***	0.008***	1

Panel B	Demand	Speed	Delivery fees	Red envelopes	Discounts	Ratings	Holiday	Weekend	Rain
Demand	1								
Speed	0.055	1							
Delivery fees	-0.027	-0.225***	1						
Red envelopes	0.133***	-0.529***	0.351***	1					
Discounts	0.097***	-0.608***	0.381***	0.807***	1				
Ratings	0.357***	0.331***	-0.223***	-0.164***	-0.229***	1			
Holiday	-0.050	-0.043	0.000	-0.028	0.005	-0.054	1		
Weekend	-0.053	0.032	-0.013	-0.025	0.001	-0.056	-0.008	1	
Rain	0.054	-0.008	0.046	-0.019	0.007	-0.020	0.143***	-0.223***	1

Note: The numbers represent correlation coefficients between variables. *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

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4.2 Distribution of order frequency and delivery fees

To gain a deeper understanding of our data, we analyze order distribution from multiple perspectives. Figure 3 displays the number of orders by day during our study period, which shows that order quantity was significantly lower during holidays and weekends, especially during the labor-day holiday (i.e., May 1st) and the following weekend (i.e., May 6th and May 7th). Order quantity increase significantly after the weekend of May 13th and 14th and reached a relatively high level afterwards, and we still observe the same pattern that the number of orders is low during weekend and holiday (i.e., May 28th-30th). The possible reason is that during holiday and weekends, people tend to eat outside for formal meals instead of ordering fast foods online.

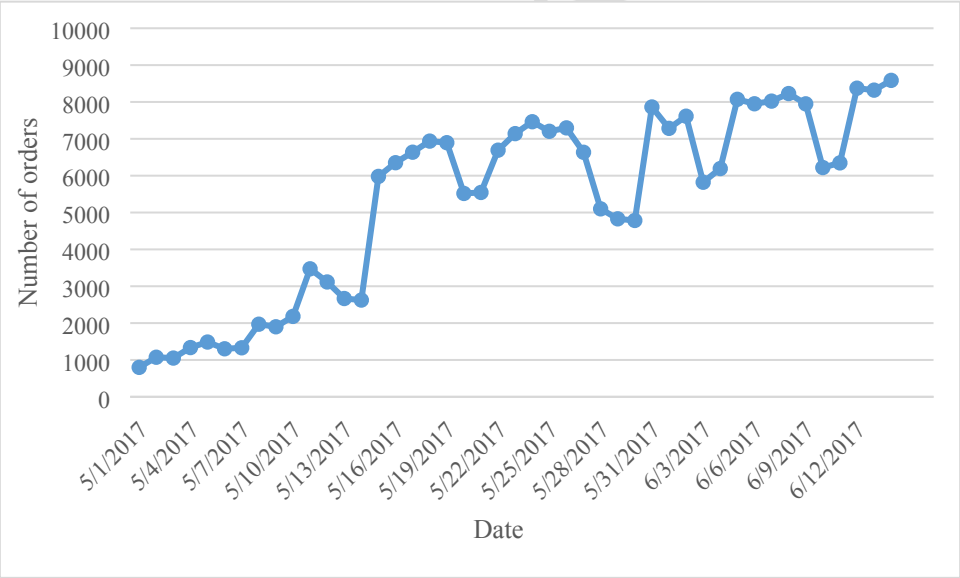

Figure 3. Distribution of order frequency by day

Figure 4 depicts the distribution of orders by hour, which is a double peak curve. The highest peak occurs at around noon from 11 AM to 1 PM, when the demand for food service online is the highest. Additionally, we also observe a smaller peak at the time period between 6 PM and 7 PM. The demand is much smaller besides the peak

hours, especially before 10 AM and after 8 PM, which we refer to as slack hours.

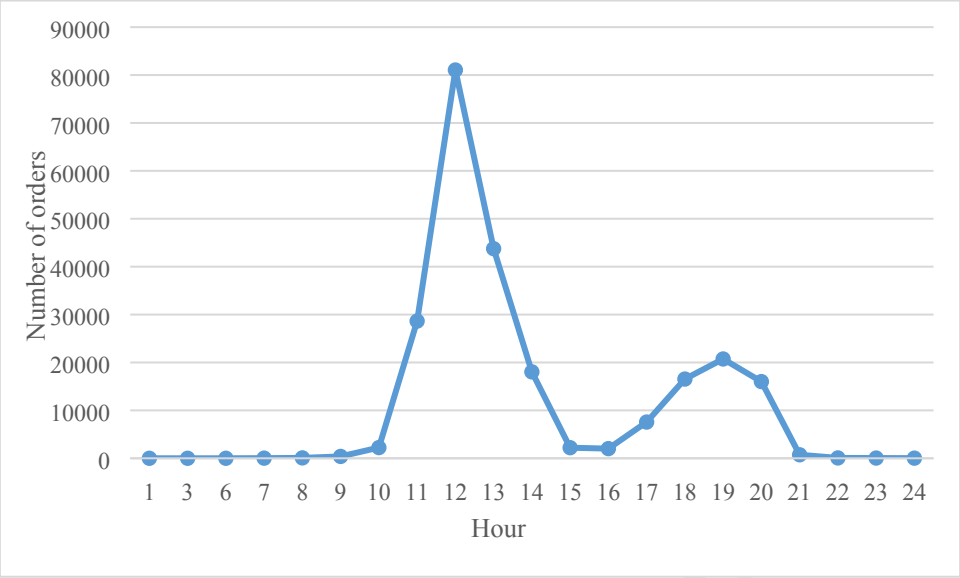

Figure 4. Distribution of order frequency by hour

Table 6 displays average delivery fees of each platform during peak hours, holidays, and weekends. Table 6 indicates that the change of average delivery fees with respect to time factors differs among different platforms. Compared with non-peak hours, M sets higher delivery fees during peak hours, while E and B set lower delivery fees during peak hours. In addition, during holidays and weekends, M and B have lower delivery fees, while E has higher delivery fees.

Table 6. Average delivery fees of different platforms

	M	E	B
Peak1 (11:00-12:00)	3.429	5.132	4.518
Peak2 (12:00-13:00)	3.420	5.137	4.580
Non-Peak	3.353	5.162	4.622
Holiday	3.334	5.173	4.296
Weekend	3.310	5.171	4.519
Rest days	3.419	5.138	4.610

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5. Empirical results

As a first step, we analyze factors affecting delivery fees. By comparing how the delivery fee of each platform changes with respect to distance, time factors (i.e., holiday, weekend, Peak1, Peak2), order price, and weather conditions, we aim to identify the pricing strategy (i.e., dynamic vs. static) of each platform. In the second step, we analyze how the pricing strategy affects the demand of each platform. Our model specifications are designed based on previous literature and real-life practices in China, which are also confirmed by a series of interviews with managers/CEOs from leading O2O companies in China.

5.1 Analysis of delivery fees

Table 7 presents the results of delivery fees analysis based on the whole sample. Model 1 displays estimation results without controlling for any fixed effects. Model 2 and 3 control for store and platform fixed effects, respectively, and Model 4 controls for two fixed effects together. The inclusion of fixed effects can reduce omitted variable bias. We also run Breusch-Pagan test to test for heteroskedasticity, and the results are summarized in Table A1 in the appendix (Breusch and Pagan 1979). Since Breusch-Pagan test indicates that variances of error terms are not constants, we report robust standard errors in all specifications to solve for heterogeneity.

Results from Model 1 to 4 are highly consistent. As expected before, longer distance will increase delivery fees. In specific, a one-kilometer increase in delivery distance increases delivery fees from 0.332 to 0.405 yuan, depending on the specification. Additionally, we find that delivery fees are significantly lower during

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holidays and weekends. On average, delivery fees during holidays are 0.103-0.118 yuan lower compared with non-holidays, and delivery fees during weekends are 0.028-0.041 yuan lower compared with weekdays. In addition, Table 7 shows that delivery fees are higher during peak hours, which is as expected because a higher delivery fee is used to alleviate the pressure on delivery capacity during peak hours. We find that during peak hours, delivery fees are 0.024-0.047 yuan higher compared with non-peak hours. Moreover, consistent with our discussions before, we find that the delivery fee is positively affected by order price and rainy weathers. A 1 yuan increase in order price will increase delivery fees by 0.001-0.007 yuan. Additionally, delivery fees are 0.125-0.16 yuan higher during rainy days compared with non-rainy days. Figure 5 shows distributions of the error term in Model 4 (i.e., the most comprehensive specification), which indicate that the error term is approximately normally distributed with zero mean.

Table 8 displays the results of delivery fees by platform. Robust standard errors are reported since Breusch-Pagan test indicates heteroskedasticity (Table A1). Models 1 to 3 display the result for M, E, and B, respectively. Table 8 shows that the impact of distance on delivery fees varies among platforms. A one-kilometer increase in distance increases delivery fees in M, E, and B by 0.394, 0.373, and 0.817 yuan, respectively. We find that B is much more sensitive to distance. The impact of distance on delivery fees in B is more than twice compared with other platforms.

Table 7. Factors affecting delivery fees

	(1)	(2)	(3)	(4)
Distance	0.332***	0.355***	0.401***	0.405***
	(0.004)	(0.004)	(0.003)	(0.003)
Holiday	-0.118***	-0.103***	-0.105***	-0.109***
	(0.009)	(0.009)	(0.006)	(0.006)
Weekend	-0.035***	-0.028***	-0.040***	-0.041***
	(0.005)	(0.005)	(0.003)	(0.003)
Peak1	0.047***	0.042***	0.037***	0.039***
	(0.005)	(0.005)	(0.003)	(0.003)
Peak2	0.036***	0.024***	0.044***	0.038***
	(0.006)	(0.006)	(0.004)	(0.004)
Order price	0.007***	0.006***	0.002***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)
Rain	0.144***	0.143***	0.130***	0.125***
	(0.006)	(0.006)	(0.005)	(0.004)
Store fixed effect	No	Yes	No	Yes
Platform fixed effect	No	No	Yes	Yes
Constant	3.411***	3.327***	5.531***	4.249***
	(0.012)	(0.019)	(0.153)	(0.152)
N	240172	240172	240172	240172
Adjusted R^2	0.086	0.198	0.599	0.657
F-statistics	1227.637	1315.886	36349.341	11743.096

Note: *, **, and *** denote significance at 10%, 5%, and 1%, respectively. Numbers in parenthesis are robust standard errors.

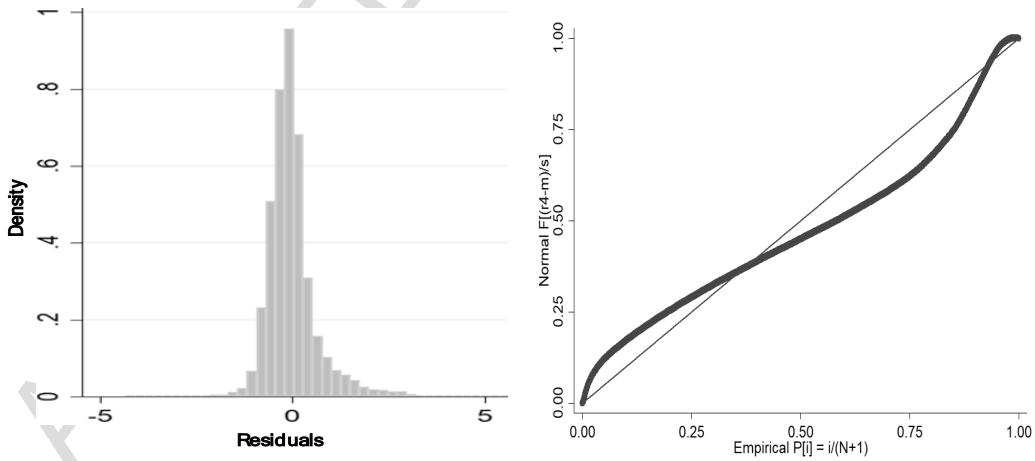

Figure 5. Distributions of the error term in Table 7 (Model 4)

Table 8 shows that delivery fees in both M and E are significantly affected by time factors, including weekends, holiday, and peak hours. However, delivery fees in B are

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not significantly affected by these time factors except for holiday. The low demand during weekends and peak demand during peak hours do not affect delivery fees charged by B. Moreover, different from other two platforms, delivery fees in B are not adjusted to rainy weathers. Overall, we find delivery fees in B depend more heavily on distance, and is not adjusted according to most time factors and weather conditions. Therefore, the pricing strategy adopted by B is less dynamic compared with other platforms, which could have an impact on its performance.

Even though both M and E adopt dynamic pricing strategies, it is interesting to note that their delivery fees adjust differently with respect to peak hours. During peak hours when the demand is high, M adjusts up its delivery fees, while E lowers delivery fees. Such results are consistent with the pattern we found in the raw data, which is displayed in Table 6. The underlying reason revealed by our theoretical model (i.e., Proposition 1) is the delivery capacity. In general, the delivery capacity of M is relatively small compared with E. As a result, during peak hours, M adjusts up delivery fees to balance drivers and consumers, while E, with sufficient delivery capacity, lowers delivery fees and thus attracts more consumers. Our interviews with the internal staff also confirm the delivery capacity differences among platforms during peak hours.

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Table 8. Factors affecting delivery fees by platform

	(1) M	(2) E	(3) B
Distance	0.394***	0.373***	0.817***
	(0.004)	(0.003)	(0.075)
Holiday	-0.175***	0.012*	-0.474***
	(0.009)	(0.007)	(0.142)
Weekend	-0.089***	0.018***	-0.054
	(0.005)	(0.004)	(0.067)
Peak1	0.109***	-0.016***	-0.034
	(0.005)	(0.003)	(0.063)
Peak2	0.101***	-0.012***	0.081
	(0.006)	(0.004)	(0.073)
Order price	0.002***	0.001***	0.010***
	(0.000)	(0.000)	(0.001)
Rain	0.210***	0.011***	0.027
	(0.007)	(0.004)	(0.092)
Constant	2.644***	4.569***	2.777***
	(0.009)	(0.005)	(0.139)
N	100115	95060	3883
Adjusted R^2	0.195	0.274	0.215
F-statistics	1867.863	2334.962	27.017

Note: *, **, and *** denote significance at 10%, 5%, and 1%, respectively. Numbers in parenthesis are robust standard errors. RMSE represents root mean square error.

5.2 Impact of dynamic pricing on demand

Table 9 displays the impact of dynamic pricing strategy on demand. Robust standard errors are reported since Breusch-Pagan test indicates heteroskedasticity (Table A1). Dynamic pricing is a dummy variable we constructed based on our estimation in the previous step, with 1 indicating M and E, and 0 indicating B. Model 1 presents the results of the overall sample, and Model 2 and 3 display the results for peak hours and slack hours, respectively.

Table 9 shows that dynamic pricing has significant impact on demand. Overall,

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platforms that adopt dynamic pricing strategy have approximately 100 more orders per hour compared with platforms that are not dynamic pricing, and such impact is even larger during peak hours (i.e., 380 orders). Even during slack hours when demand is very low, dynamic pricing is shown to increase the number of orders by approximately 9 per hour. The positive impact of dynamic pricing on demand is consistent with Proposition 2 in our theoretical model, which indicates that compared with static pricing strategies, dynamic pricing strategies yield better performance.

In addition to the pricing strategy, during certain time periods, demand is also affected by the delivery speed, delivery fees, red envelopes, ratings, weather, holidays, and weekends. For instance, we find delivery speed has positive and significant impact on demand during peak hours, which is as expected because time is more valued during peak hours. Specifically, if speed increases by one minute/kilometer, the number of orders per hour will increase by approximately 36 during peak hours. Table 9 also shows that red envelopes will increase demand significantly, especially during peak hours. In specific, a one yuan increase in red envelopes will increase hourly demand by 188 during peak hours. Moreover, Model 1 suggests that if ratings increase by one (out of five), the number of orders per hour will increase by approximately 116.

Moreover, Table 9 shows that delivery fees only affect demand during slack hours, suggesting that consumers are more sensitive to delivery fees during non-regular meal times. Moreover, consistent with previous findings, we find demand is higher during rainy days and lower during holidays and weekends. Figure 6 presents distributions of the error term of Model 1, which indicate that the error term is approximately normally

distributed with zero mean.

Table 9. The impact of dynamic pricing on demand

	(1) Overall	(2) Peak hours	(3) Slack hours
Dynamic pricing	99.586***	379.722***	9.229***
	(14.178)	(103.150)	(1.590)
Speed	-0.532	36.052***	0.046
	(0.796)	(6.787)	(0.065)
Delivery fees	5.379	70.857	-1.424***
	(5.447)	(57.008)	(0.399)
Red envelopes	18.923	187.941***	-0.110
	(11.980)	(69.839)	(0.625)
Discounts	6.450	36.945	-0.131
	(5.802)	(44.940)	(0.339)
Ratings	116.066***	-48.706	0.913
	(17.225)	(43.604)	(0.824)
Rain	50.394**	108.824	1.334
	(21.339)	(67.144)	(2.561)
Holiday	-36.676***	-155.454***	-1.041
	(13.938)	(53.309)	(2.561)
Weekend	-11.808	-118.078***	0.446
	(11.308)	(43.361)	(1.766)
Constant	-0.415	-367.550**	7.417*
	-24.079	-171.351	-4.105
N	1564	268	274
Adjusted R^2	0.185	0.487	0.056
F-statistics	66.837	46.602	7.224

Note: *, **, and *** denote significance at 10%, 5%, and 1%, respectively. Numbers in parenthesis are robust standard errors.

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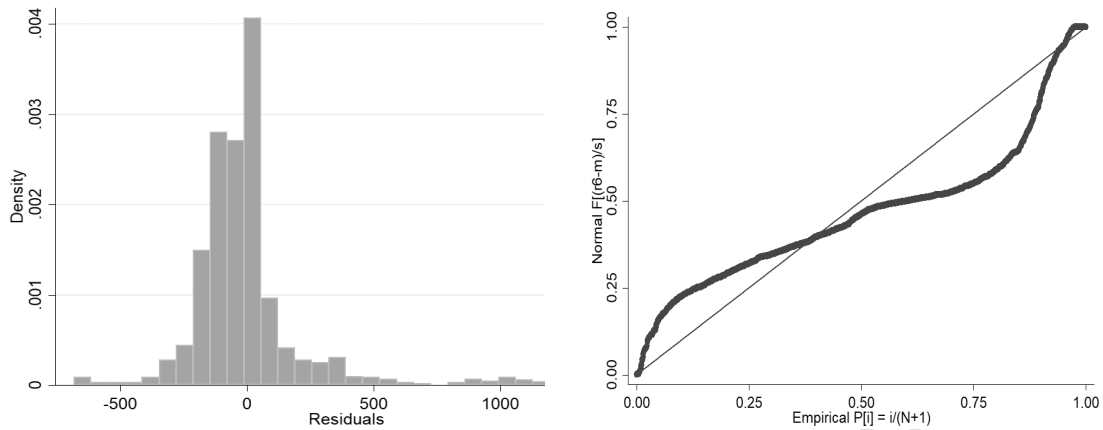

Figure 6. Distributions of the error term in Table 9 (Model 1)

6. Conclusion and discussion

6.1 Conclusions

This study identifies the pricing strategy of three most-commonly used on-demand food service platforms (i.e., M, E, and B) in China, and further evaluates the impact of pricing strategies on platform performance. Based on the distinct market structure of on-demand food service market in China, we develop a dyadic two-sided market theory, in which the platform connects not only merchants and customers, but also merchants and drivers. Our theoretical model demonstrates that dynamic pricing strategies outperform static pricing strategies.

Based on detailed information of 240,172 orders, our empirical analysis reveals that M and E adopt dynamic pricing strategies, while B adopts static pricing strategies. Consistent with the theoretical model, we find that platforms adopting dynamic pricing strategies have a significantly higher level of demand compared with platforms adopting static pricing strategies. Specifically, platforms that adopt dynamic pricing strategies have approximately 100 more orders per hour compared with platforms that

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adopt static pricing strategies, and such impact is even larger during peak hours. In addition, we find attributes such as delivery speed, delivery fees, red envelopes, weather, and holiday and weekends all significantly impact demand.

6.2 Managerial implications

Based on empirical results, this study has several important managerial implications. First, we suggest the type of pricing strategy that platforms should adopt to improve performance. In addition, we make several suggestions about the usage of red envelopes, how to improve ratings and delivery speed, and how to balance supply and demand during extreme weather conditions, holidays, and weekends.

Specifically, we suggest that on-demand food service platforms in China should adopt a dynamic pricing strategy to improve performance. Dynamic pricing strategies have been proved to be an effective means to improve performance of e-commerce (Cachon et al. 2017, Hall et al. 2015). Surge pricing is useful to balance supply and demand by pricing out consumers who have lower willingness to pay while attracting more workers and reducing the shortage in supply (Guda and Subramanian 2017). In addition, the specific dynamic pricing strategy should be designed based on the delivery capacity of the platform. If there are not enough delivery personnel during peak hours, platforms should adjust the delivery fee upward to attract more drivers to meet the increasing demand. In contrast, if the delivery capacity is sufficient during peak hours, reducing the delivery fee can be an effective strategy to reduce the platform's cost.

Under limited budgets and resources, platforms should give the highest priorities to improve the most important determinant that affects demand. Specifically, platforms

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should offer more red envelopes during peak hours, which can increase demand significantly. The importance of red envelopes can be reflected by promotion strategies adopted by many leading on-demand food service platforms. For instance, both Meituan takeout and Eleme in China have various forms of red envelopes. Both platforms send red envelopes to potential customers who logged in their APPs and customers who placed orders. Developing new forms of red envelopes and increasing the number of red envelopes during peak hours can stimulate customers place more orders and improve the performance of platforms.

Platforms should put more efforts into improving ratings. With the rapid development of information technology, ratings serve as electronic word of mouth (eWOM) to influence customers' perceptions and behaviors (Cantallops and Salvi 2014, Xu 2018). The ratings posted by previous customers affect future customers' order decisions. A good rating serves as a credible confirmation of the service provided by the platform and thus increases the attractiveness of the platform (Zhang et al. 2010). To amplify positive eWOM, platforms can make full use of social media by creating online discussion forums, building online service quality management teams, and investing more in advertisement. It is also necessary to offer prompt responses to online complaints by explaining reasons for service failures and offering reasonable compensations (Xu, 2018).

Platforms need to invest more resources to increase the delivery speed during peak hours. Order fulfillment can be the most expensive and critical operation for businesses of companies engaged in e-commerce. An e-tailer's success depends on its ability to

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fulfill and deliver orders on time to a large extent (Stank et al. 1999, Lee and Whang 2001). On-demand food service industry typically has high mobility rates because it does not require professional knowledge and skills for drivers to enter the industry. To develop and sustain high-performance delivery teams, platforms should offer more benefits to drivers. For instance, platforms can set up a motivation system, in which drivers receive higher bonuses when they deliver more orders.

Finally, considering the demand is higher during rainy weather and lower during holidays and weekends, platforms can invest more resources to balance supply and demand and guarantee the on-time delivery during rainy days and stimulate demand on holidays and weekends. For instance, to alleviate delivery pressure during rainy days, platforms can increase delivery fees, reduce delivery radius, and reduce the number of promotional activities.

6.3 Limitations and extensions

The limitations of this study mainly lie in the following aspects. First, in our theoretical model, assumptions 3 and 4 will not hold if platforms require merchants to use their delivery service in the future. In this case, the utility of merchants will depend on the number of platform drivers in addition to the number of customers. Future studies can test the robustness of results by relaxing these assumptions. In addition, the on-demand food service market has gone through significant changes during the second half of 2017 and the year of 2018. Baidu takeout has become part of Eleme, but still operated independently with the same operation system and personnel structure. Didi made a strong entry to the market on March 2018. It would be interesting if we could have more

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recent data to reflect the changes in the market. Future studies can address this limitation by collecting more data from more on-demand food delivery platforms, and identifying and comparing the pricing strategy with different types of platforms (e.g., old versus newly established platforms). Moreover, in addition to food service market evaluated in the current study, on-demand services span a wide range of products such as fresh groceries and medicines. With significant differences in the customer pool and peak hours of demand across various on-demand services platforms, it would be interesting to extend our analysis to other types of markets in future studies.

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Appendix

Table A1 Summary of Results of Breusch-Pagan test

Table 7	Model 1	Model 2	Model 3	Model 4
Chi^2	23220	15421	1985	2332
Probability	0	0	0	0
Table 8	Model 1	Model 2	Model 3	
Chi^2	23354	35640	3165	
Probability	0	0	0	
Table 9	Model 1	Model 2	Model 3	
Chi^2	1139	46.95	18.12	
Probability	0	0	0	

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